A Faster Local Correction and Quantitative Logarithmic Control for Littlewood Lower Envelopes

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Abstract

We examine the specific sampling, gap correction, and rounding architecture of *Nearly minimal maxima and positive minima of Littlewood polynomials*. For fixed auxiliary geometry, an acute-angle argument allows taper width $c_\delta N^{-1/2}$: both the lower-envelope loss and the discarded Fourier mass become $O_\delta(N^{-1/2})$, replacing the source's $O_\delta(N^{-1/4})$ projection loss without enlarging its coefficient budget. We also determine the sharp worst-case rounding defect from the available second moment and coefficient perturbation, and propagate it through the actual normalization. A fixed positive floor yields explicit sharp Jensen bounds, exceptional-set estimates, and quantitative logarithmic control; separated products are a secondary application, not a dynamical realization. All existence statements using the upstream spreading and packing construction remain conditional on those inputs. Neither effective headline thresholds nor a resolution of conflicting public flatness claims is asserted. Contribution assessment: C3/5, a source-specific improvement of the local correction rate and a sharp rescaling budget, with global construction and dynamical obligations explicitly retained.

1 Normalization, scope, and dependency map

Throughout, $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ has probability Haar measure dt and $\mathrm{e}(t) = \exp(2\pi i t)$. For $X \in \mathbb{R}^N$ put

$$U_X(t) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X_k \mathrm{e}(kt), \qquad \mu(X) = \frac{1}{2} \sum_{k=0}^{N-1} (1 - |X_k|) \quad (|X_k| \le 1).$$

The degree is $N-1$, not N. For signs $\varepsilon_k \in \{-1, 1\}$, Parseval gives $\int |U_\varepsilon|^2 = 1$ exactly. Fourier coefficients are $\widehat{g}(k) = \int_{\mathbb{T}} g(t)\mathrm{e}(-kt)\, dt$.

The source claim is: for every fixed $\eta > 0$ there is $N_0(\eta)$ such that for every integer $N \ge N_0(\eta)$ some sign polynomial satisfies

$$\frac{1}{16} \le |U_\varepsilon(t)| \le 1 + \eta \qquad (t \in \mathbb{T}). \tag{1}$$

This report does not replace its $1/16$ by an asymptotically unit lower bound, or use a companion manuscript as evidence. The following dependency map records what was actually inspected. Line numbers refer to the cached `build/sections/` files, all read in full; the precise read batches and pre/post hashes are in the owned read ledger.

File and lines	Role and retained obligation
`spreading`, 1–408	Bounded real auxiliary polynomial, nonparallel Fourier pairs, stationary-phase coefficients and an ℓ^1 truncation. Auxiliary dimensions and constants depend on fixed δ.
`packing`, 1–307	Finite-field angle slots and a nearly regular hypergraph matching. The matching theorem is imported; dependencies among three or more forms are not replaced by independence.
`sampling`, 1–334	Poisson/Fresnel sampling at every sufficiently large integer length; disjoint frequency intervals and a fixed modulus profile with short gaps.
`correction`, 1–407	Piecewise quadratic gap waves, Hermitian reflection, coefficient budget and exterior Fourier tail. This is the main local improvement here.
`rounding`, 1–236	Lovett–Meka partial coloring, mass-controlled dyadic rounding and grid-to-circle comparison. The partial-coloring theorem remains an imported input.
`completion`, 1–85	Rescaling, defect estimate and order of limits: choose δ first, then N.

The introduction (146 lines), main file, bibliography, figure source and README were also read. Literal reading is not an independent certification of every imported theorem or of the global claim. In particular, our conditional assembly below isolates the sampling output rather than claiming a new unconditional construction.

1.1 The specific sampling interface

Write $b = 1/8$, $A = 1/2$, and

$$K_\delta = \sqrt{\frac{(1+\delta)^3}{1-\delta}}.$$

For fixed sufficiently small $\delta > 0$, the source supplies real $Y_{N,k} \in [-1, 1]$ and a Hermitian-symmetric B_N such that

$$\|U_Y - B_N\|_\infty \le C_\delta N^{-1}, \qquad \frac{1}{N}\sum_k Y_{N,k}^2 \ge 1 - 8\delta. \tag{2}$$

The smooth even modulus $B = |B_N|$ is independent of N, is at most K_δ, and vanishes near 0 and $1/2$. The gaps $J_j = (u_j, v_j) \subset [0, 1/2]$ where $B < b$ form a fixed finite family, of total length $L \le 10\delta$. At each transition endpoint z,

$$B_N(t) = B(t)\mathrm{e}(\alpha_z + N\psi_z(t)), \qquad B(z) = b, \qquad \psi_z'(z) \in (0, 1), \tag{3}$$

on a fixed neighborhood; ψ_z is quadratic. Boundary gaps at 0 and $1/2$ use reflection instead of a transition endpoint.

These premises are stronger than just a mean-square estimate. For context, the source first constructs a bounded real multivariate polynomial F with $\|F\|_2 \ge 1 - 3\delta$ and Fourier support $\{\pm a_1, \dots, \pm a_r\}$, with no parallel pairs other than opposites. Its velocity v obeys

$$\lambda_a = a \cdot v \ne 0, \quad \sum_a |\lambda_a| < 1, \quad \frac{|\widehat{F}(a)|}{\sqrt{|\lambda_a|}} \le K_\delta.$$

Angle packing separates all stationary intervals; distinct λ_a are not required. Poisson summation converts a quadratic sampled phase to its Fresnel main term uniformly in the circle variable. Disjoint supports give $\int B^2 \ge 1 - 7\delta$, and comparison with $B \le K_\delta$ gives $L \le 10\delta$. We do not infer these strong premises merely from a Fourier change of coordinates.

2 Acute-angle gap correction

We retain the source's fixed finite geometry, but change its shrinking taper. Here and below C_δ may depend on the actual gap lengths, slot separations, phase derivatives, and cutoff profiles. It is not uniform as $\delta \downarrow 0$.

2.1 Geometry and variable taper

Let $\ell_j = v_j - u_j$. Choose disjoint closed derivative slots $D_j = [p_j, q_j] \subset (1/4, 3/4)$ with width $\ell_j/(4L)$. On each gap, the source chooses a continuous piecewise affine derivative ξ_j which goes from the prescribed endpoint slope to p_j, then to q_j, then to the other endpoint slope. Its two outer pieces are fixed and sufficiently short that

$$\sum_{\text{outer pieces}} |\xi_j'|^{-1/2} \le \sqrt{\delta}, \qquad \xi_j' \ge (4L)^{-1} \quad \text{on middle pieces.} \tag{4}$$

The primitives ϕ_j are C^1 and piecewise quadratic, with $c_\delta \le \phi_j' \le 1 - c_\delta$. At transition endpoints $\phi_j'(z) = \psi_z'(z)$; at the reflection endpoints the slope is $1/8$. An affine $\gamma_{j,N}$ aligns endpoint phases modulo one and obeys

$$|\gamma_{j,N}'| \le \ell_j^{-1}, \qquad \int_{J_j} |\gamma_{j,N}'| \le 1. \tag{5}$$

Its endpoint values can be chosen in $[0, 1)$. At 0 and $1/2$ the complete correction phase is an integer.

For a smooth step ϑ from zero to one on $[0, 1]$, multiply a step $\vartheta((t - u_j)/w)$ or $\vartheta((v_j - t)/w)$ for each transition endpoint, and none for a reflection endpoint. Their product is $r_{j,N}$. Whenever w is smaller than all fixed outer-piece and phase-neighborhood lengths, these strips are disjoint and

$$0 \le r_{j,N} \le 1, \quad \int |r_{j,N}'| \le 2, \quad \sup |r_{j,N}'| + \int |r_{j,N}''| \le C_\delta w^{-1}. \tag{6}$$

Set

$$R_N(t) = A r_{j,N}(t) \mathrm{e}(N\phi_j(t) + \gamma_{j,N}(t)) \quad (t \in J_j), \tag{7}$$

zero elsewhere on the positive half-circle, and $R_N(-t) = \overline{R_N(t)}$. It is continuous: it vanishes at transition ends, and equals A at both reflection joins.

Theorem 2.1 (Fixed-geometry improved correction)**.** *Under the above premises there is a fixed $\kappa_\delta > 0$ such that, with $w = \kappa_\delta N^{-1/2}$, all sufficiently large N satisfy*

$$b - C_\delta N^{-1/2} \le |B_N(t) + R_N(t)| \le K_\delta \qquad (t \in \mathbb{T}), \tag{8}$$

$$\sqrt{N}\, |\widehat{R}_N(k)| \le D\sqrt{\delta} \qquad (k \in \mathbb{Z}), \tag{9}$$

$$\sum_{k<0 \text{ or } k \ge N} |\widehat{R}_N(k)| \le C_\delta N^{-1/2}. \tag{10}$$

Here D is absolute, after enlarging the δ-dependent threshold. All $\widehat{R}_N(k)$ are real. Consequently the consecutive-frequency projection $\Pi_N R_N = \sum_{k=0}^{N-1} \widehat{R}_N(k) \mathrm{e}(kt)$ differs uniformly from R_N by $O_\delta(N^{-1/2})$.

Proof. At a transition endpoint subtract the common endpoint phase modulo an integer. Equality of the leading first derivatives and their fixed Lipschitz bounds give, on the strip, a phase difference in turns bounded by

$$|\Delta(t)| \le C_\delta(Nw^2 + w). \tag{11}$$

Choose κ_δ with $C_\delta\kappa_\delta^2 \le 1/16$, and then N large enough that $C_\delta\kappa_\delta N^{-1/2} \le 1/16$, as well as the containment conditions for every strip. Thus $|\Delta| \le 1/8$. For real $B, r \ge 0$,

$$|B + Are(\Delta)|^2 - B^2 = A^2r^2 + 2BAr\cos(2\pi\Delta) \ge 0. \tag{12}$$

Rotating by the phase of B_N therefore proves $|B_N + R_N| \ge B(t) \ge b - \|B'\|_\infty w$ on the strip. Away from strips, in a gap, $r = 1$ and the lower bound is $A - b = 3/8$. Outside gaps it is $B \ge b$. On gaps the upper bound is $b + A = 5/8$, and elsewhere K_δ. Reflection preserves these estimates. Notice that no convergence of Δ to zero was used.

On a positive quadratic piece write $a = Are(\gamma)$ and $\phi = \phi_j$; on its reflection use $a(t) = \overline{a(-t)}$, $\phi(t) = -\phi_j(-t)$. The reflected derivative is $\xi_j(-t)$, so the same slot is used. Equations (5)–(6) imply the width-independent bound

$$\sup|a| + \int |a'| \le A + 2A + 2\pi A =: C_1. \tag{13}$$

Completing the square bounds the primitive of $e(N\phi - kt)$ by $C(N|\phi''|)^{-1/2}$ if $\phi'' \ne 0$: after scaling, the central Fresnel integral is bounded by its length, and each exterior half-line by integration by parts with derivative proportional to its coordinate. Integrating against this primitive gives the same bound times (13). A monotone phase derivative of absolute value at least $N\rho$ similarly gives $CC_1/(N\rho)$.

The outer pieces and their reflections contribute at most $2C_2\sqrt{\delta}/\sqrt{N}$ by (4). Choose fixed $\rho > 0$ such that the ρ-neighborhoods of the slots are disjoint. At most one middle slot is close to k/N, and it and its reflection contribute at most $4C_2\sqrt{L}/\sqrt{N}$. All other pieces contribute $O_\delta(N^{-1})$. Therefore

$$\sqrt{N}|\widehat{R}_N(k)| \le (2 + 4\sqrt{10})C_2\sqrt{\delta} + O_\delta(N^{-1/2}),$$

uniformly in k. Take $D = (2+4\sqrt{10})C_2 + 1$. The bound did not change with taper width. Hermitian symmetry and the change of variable $t \mapsto -t$ give $\overline{\widehat{R}_N(k)} = \widehat{R}_N(k)$.

For an exterior integer k, put $T = N + |k|$ and $V = N\phi' - k$. On each piece

$$|V| \ge c_\delta T, \quad |V'| \le C_\delta N, \quad V'' = 0, \qquad \sup|a'| + \int |a''| \le C_\delta w^{-1}.$$

The last estimate follows because the w^{-2} second derivative of a step is supported on a set of length w; the affine γ has fixed derivative bounds. One integration by parts gives

$$\int_I a e(N\phi - kt) = \left[\frac{a e(N\phi - kt)}{2\pi i V}\right]_{\partial I} - \frac{1}{2\pi i}\int_I (a/V)' e(N\phi - kt).$$

The first boundary terms cancel in the sum: $R_N e(-kt)$ and ϕ' agree at internal joins; the numerator vanishes at transition ends; at zero the numerator is A and both derivatives are $1/8$; at the identified endpoints $\pm 1/2$ the numerator is $A(-1)^k$ and the derivatives again agree. No cancellation after the next integration is assumed. Set

$$Q = \frac{a'}{V^2} - \frac{aV'}{V^3}, \qquad Q' = \frac{a''}{V^2} - \frac{3a'V'}{V^3} + \frac{3a(V')^2}{V^4}.$$

The second boundary terms and integrals are bounded, in total, by

$$|\widehat{R}_N(k)| \le C_\delta\left(\frac{w^{-1}}{T^2} + \frac{N}{T^3} + \frac{N^2}{T^4}\right) \le \frac{C_\delta w^{-1}}{(N + |k|)^2}. \tag{14}$$

Summation gives $C_\delta/(Nw) = O_\delta(N^{-1/2})$. The Fourier series is absolutely convergent; its continuous sum equals R_N by Fourier uniqueness. Its discarded tail bounds the uniform projection error. □

Remark 2.2 (What has improved). With $w = N^{-\alpha}$ and the source's small-complex-error estimate, the two losses are $N^{1-2\alpha}$ and $N^{\alpha-1}$, for $1/2 < \alpha < 1$. Choosing $\alpha = 2/3$ balances them at $N^{-1/3}$, instead of the source's bottleneck $N^{-1/4}$ at $\alpha = 3/4$. The acute-angle inequality is stronger: it permits a fixed small phase error and reaches $N^{-1/2}$ for both the envelope and projection. This is not a claim of an optimal rate over all possible corrections, or an effective estimate for $N_0(\eta)$ in (1).

3 Sharp rescaling defect and the rounding interface

Proposition 3.1 (Exact worst-case defect)**.** *Let $0 \le s < 1$, $Y, r \in \mathbb{R}^N$, $|Y_k| \le 1$, $|r_k| \le s$, and $N^{-1}\sum Y_k^2 \ge M \in [0,1]$. For $X = (Y+r)/(1+s)$, put $q(X) = \mu(X)/N$. Then*

$$q(X) \le q_*(M,s) := \frac{1}{2}\left(1 - \frac{(M-s^2)_+}{(1+s)^2}\right). \tag{15}$$

This is sharp over probability distributions of the coordinates, and over finite vectors whenever the relevant two-point proportion is rational with denominator dividing N.

Proof. For $0 \le t \le 1$,

$$(t-s)_+ \ge \frac{t^2 - s^2}{1+s}.$$

For $t < s$ the right side is negative; for $t \ge s$ the difference is $(t-s)(1-t)/(1+s) \ge 0$. Thus

$$\frac{1}{N}\sum |X_k| \ge \max\left(0, \frac{N^{-1}\sum Y_k^2 - s^2}{(1+s)^2}\right),$$

which proves the bound. If $M \ge s^2$, put mass $p = (M-s^2)/(1-s^2)$ at $Y = 1$ and the remainder at $Y = s$, always taking $r = -s$. Equality holds at both points. If $M \le s^2$, take $Y = \sqrt{M}$ and $r = -Y$, so $X = 0$. Irrational p can be approximated by empirical proportions; this sharpness is for the stated information class, not a claim that the Fourier correction attains its extremizer. □

In the source assembly set $s = D\sqrt{\delta}$, $M = 1 - 8\delta$, and choose δ so $0 < s < 1$ and $M > s^2$. Then

$$q_\delta^* = \frac{s + s^2 + 4\delta}{(1+s)^2} < \frac{1}{2}, \qquad q(X) \le q_\delta^* \le 4\delta + 2s. \tag{16}$$

The last term is the source budget. The intermediate bound $(4\delta + s)/(1+s)$ follows already from a first-moment triangle inequality; (16) is smaller still by $4\delta s/(1+s)^2$. It improves the leading coefficient of the s loss from 2 to 1, but does not remove the $\sqrt{\delta}$ correction scale.

3.1 Why the source rounding input has this form

We retain the source's defect-sensitive consequence of partial coloring: there is an absolute C such that every real $X \in [-1,1]^N$ admits signs with

$$\|U_\varepsilon - U_X\|_\infty \le \frac{C}{\sqrt{N}} + C\sqrt{q(X)\log\frac{80}{q(X)}}. \tag{17}$$

At $q = 0$ the square-root term is zero. The imported input is the Lovett–Meka partial-coloring statement [3]; the following explains the source's transformations, rather than treating (17) as a consequence of Parseval.

For a real $R \times s$ matrix with entries in $[-1,1]$, $s \le R$, successive partial colorings of at most $2^{-h}s$ uncolored variables give discrepancy at most $C_0\sqrt{s\log(2R/s)}$. The exact-half existence assertion follows from the approximate version by compactness in the fixed finite cube; this is not a limiting polynomial-time claim. Write $p_k = (1-|X_k|)/2$ and $X_k = \sigma_k(1-2p_k)$, $\sigma_k \in \{-1,1\}$. Use real cosine and sine rows on the $20N$-point circle grid, so $R = 40N$. After dyadic truncation with $N2^{-J} \le 1$, the odd numerators at step j involve at most $\min(N, 2^j\mu(X))$ variables. Reversing all signs of a discrepancy solution, if needed, makes the total mass nonincreasing. The intermediate dyadic coordinates remain in $[0,1]$; they need not remain in $[0,1/2]$. The sum of step errors is bounded by

$$1 + \frac{C_0}{\sqrt{2}-1}\sqrt{\mu(X)\log\frac{80N}{\mu(X)}}.$$

Combining cosine and sine costs at most $2\sqrt{2}$ before normalization. Finally, if a degree-$(N-1)$ polynomial has circle supremum S, Cauchy's estimate on the radius $1+1/N$ circle bounds its t derivative by $2\pi eNS$. Nearest-grid distance is at most $1/(40N)$, so $S \le G + (\pi e/20)S$ and $S \le G/(1-\pi e/20)$. This accounts for the continuous, rather than just sampled, norm.

Corollary 3.2 (Fixed-parameter finite assembly)**.** *Assume* (2)*, the correction premises and* (17)*. For fixed sufficiently small δ as above, there are signs such that, for all sufficiently large N,*

$$|U_\varepsilon(t)| \ge \frac{b}{1+s} - C\sqrt{q_\delta^*\log(80/q_\delta^*)} - C_\delta N^{-1/2}, \tag{18}$$

$$|U_\varepsilon(t)| \le \frac{K_\delta}{1+s} + C\sqrt{q_\delta^*\log(80/q_\delta^*)} + C_\delta N^{-1/2}. \tag{19}$$

Both hold for every $t \in \mathbb{T}$. Consequently any fixed lower constant $c < 1/8$ and any fixed upper constant $1+\eta > 1$ are obtained if the same upstream construction is valid, by choosing δ and then N.

Proof. Take $r_k = \sqrt{N}\widehat{R}_N(k)$ and $X = (Y+r)/(1+s)$. Then $|X_k| \le 1$ and

$$U_X = \frac{B_N + R_N}{1+s} + O_\delta(N^{-1/2})$$

uniformly, using sampling and the improved projection tail. Apply Proposition 3.1 and (17); the function $q\log(80/q)$ increases on $[0,1/2]$. As $\delta \to 0$, $s, q_\delta^* \to 0$ and $K_\delta \to 1$. This proves the last assertion with strict margins before increasing N. □

The source completion already leaves room for every fixed $c < 1/8$; that observation is not credited as a new existence theorem. The new information is the $N^{-1/2}$ fixed-geometry remainder and the smaller explicit defect budget. Large auxiliary dimensions, gap counts and slot separation constants still obstruct a useful uniform headline threshold.

4 What the fixed floor adds

Let (Ω, ν) be a probability space and assume

$$0 < a < 1 < b, \qquad a \le f \le b \quad \nu\text{-a.e.}, \qquad \mathbb{E}f = 1. \tag{20}$$

For the source claim, $f = |U_\varepsilon|^2$, $a = 1/256$ and $b = (1+\eta)^2$. Keeping a fixed is essential to the uniform logarithmic constants below. Define

$$\tau = \frac{(b-1)(1-a)}{b-a}, \qquad J_a(b) = \frac{-(b-1)\log a - (1-a)\log b}{b-a}.$$

Proposition 4.1 (Endpoint budgets)**.** *Under* (20)*,*

$$D := \mathbb{E}(f-1)_+ = \mathbb{E}(1-f)_+ \le \tau, \qquad \mathbb{E}|f-1| \le 2\tau, \tag{21}$$

$$\operatorname{Var} f \le (b-1)(1-a), \qquad 0 \le -\mathbb{E}\log f \le J_a(b), \tag{22}$$

$$\mathbb{E}|\log f| \le \left(1 + \frac{-\log a}{1-a}\right)\tau. \tag{23}$$

If $ab \le 1$*, the sharper, sharp probability-class bound is*

$$\mathbb{E}|\log f| \le L_a(b) := \frac{-(b-1)\log a + (1-a)\log b}{b-a}. \tag{24}$$

The bounds for D, variance and Jensen gap are sharp in the same class. They do not assert that a Littlewood modulus realizes a two-point law.

Proof. Convexity gives $(x-1)_+ \le (b-1)(x-a)/(b-a)$ for $a \le x \le b$. Taking expectations proves (21). The nonnegative polynomial $(f-a)(b-f)$ has expectation at least zero, giving $\mathbb{E}f^2 \le a+b-ab$ and the variance bound. Concavity of log gives its affine lower chord between a and b; averaging gives $\mathbb{E}\log f \ge -J_a(b)$. Jensen's inequality gives the other direction. On $[a,1]$, the convex function $-\log x$ lies below its endpoint chord $(-\log a)(1-x)/(1-a)$, while on $[1,b]$, $\log x \le x-1$. Adding the resulting integrals gives (23).

For (24), let ℓ be the affine line through $(a, -\log a)$ and $(b, \log b)$. The condition $ab \le 1$ makes its slope nonpositive. On $[1,b]$, $\ell(x) \ge \log b \ge \log x$. On $[a,1]$, $\ell(a) = -\log a$ and $\ell(1) \ge 0$; hence ℓ lies above the chord of $-\log x$ on $[a,1]$, and thus above $-\log x$. Average $|\log f| \le \ell(f)$ to obtain the formula. All stated sharp bounds are equalities for the law assigning masses $(b-1)/(b-a)$ to a and $(1-a)/(b-a)$ to b. □

For fixed a, all the right sides in (21)–(24) are $O_a(b-1)$. At the source floor, for $0 < \eta \le 1$, the condition $ab \le 1$ holds. In particular,

$$\lim_{b\downarrow 1} \frac{J_a(b)}{b-1} = \frac{-\log a - (1-a)}{1-a}, \qquad \lim_{b\downarrow 1} \frac{L_a(b)}{b-1} = 1 + \frac{-\log a}{1-a}.$$

These are fixed-floor constants, not assertions of uniform pointwise flatness. Classical Jensen and convex-chord inequalities are the tools; the contribution here is their explicit accounting at the source's $a = 1/256$ normalization.

Corollary 4.2 (Exceptional sets, Mahler measure, and correlations)**.** *For* $a \le c < 1$ *and* $0 < t \le 1-a$*,*

$$\nu\{f \le c\} \le \frac{b-1}{b-c}, \qquad \nu\{f \le 1-t\} \le \min\left(\frac{\tau}{t}, \frac{b-1}{b-1+t}\right). \tag{25}$$

For a sign polynomial satisfying the envelope,

$$e^{-J_a(b)/2} \le \frac{\mathcal{M}(P)}{\sqrt{N}} \le 1, \qquad \mathcal{M}(P) = \exp\int_{\mathbb{T}} \log|P(\mathrm{e}(t))|\,dt. \tag{26}$$

If $C_r = \sum_{k=0}^{N-1-r} \varepsilon_k \varepsilon_{k+r}$*, then*

$$\frac{2}{N^2}\sum_{r=1}^{N-1} C_r^2 = \operatorname{Var} f \le (b-1)(1-a). \tag{27}$$

Proof. Writing $\alpha = \nu\{f \le c\}$ gives $1 = \mathbb{E}f \le \alpha c + (1-\alpha)b$. The other exceptional estimate is Markov applied to $(1-f)_+$. Formula (26) follows from $\log(|P|/\sqrt{N}) = \frac{1}{2}\log f$. Finally,

$$f(t) = 1 + \frac{1}{N}\sum_{r=1}^{N-1} C_r(\mathrm{e}(rt) + \mathrm{e}(-rt));$$

Parseval yields (27) with the factor 2 shown. □

For $N \ge 2$, $C_{N-1} = \pm 1$, so the usual merit factor $N^2/(2\sum C_r^2)$ is well-defined and would be at least $1/((b-1)(1-a))$. This is a conditional consequence of the envelope, not an independent construction of large merit factors.

5 A secondary logarithmic product consequence

The following application is included to delimit the relevance of the private spectral arguments. It is not credited as a new general Riesz product theorem, and does not require or supply a measure-preserving map.

Proposition 5.1 (Summable log budget with exact finite normalization)**.** *Suppose actual sign polynomials P_j of lengths $N_j \ge 2$ satisfy $a \le f_j = |P_j|^2/N_j \le b_j = 1+\epsilon_j$ on $\mathbb{T}$, for a fixed $0 < a < 1$, with $\sum_j \epsilon_j < \infty$. Choose positive integers d_j such that*

$$d_j > \sum_{i<j} d_i(N_i - 1), \qquad W_n(t) = \prod_{j=1}^{n} f_j(d_j t). \tag{28}$$

Then $\int W_n = 1$ exactly and W_n converges almost everywhere and in every finite L^p to a density W with $0 < W \le B_* := \prod_j b_j < \infty$ almost everywhere and $\int W = 1$. Moreover,

$$\|\log W - \log W_n\|_1 \le \left(1 + \frac{-\log a}{1-a}\right)\sum_{j>n} \frac{\epsilon_j(1-a)}{b_j - a}, \tag{29}$$

$$0 \le -\int \log W = \sum_j \left(-\int \log f_j\right) \le \sum_j J_a(b_j), \tag{30}$$

$$\|W - W_n\|_1 \le 2B_* \sum_{j>n} \frac{\epsilon_j(1-a)}{b_j - a}. \tag{31}$$

No positive essential lower bound for W is asserted.

Proof. The Fourier support of f_j is in $[-(N_j-1), N_j-1]$ and its constant coefficient is one. In any nonzero choice of frequencies, the last nonzero term $d_j k_j$ has absolute value at least d_j, greater than the total possible preceding magnitude. Hence no such choice sums to zero, proving finite normalization. This is not independence of the random variables $f_j(d_j t)$.

Integer dilation preserves Haar measure. By (23) and Tonelli, $\sum_j |\log f_j(d_j t)| < \infty$ almost everywhere and in L^1. Its exponential is finite and strictly positive, and proves (29) and (30). Since $W_n \le B_*$, dominated convergence yields mass one and convergence in every finite L^p. Telescoping finite products gives $\|W_m - W_n\|_1 \le B_* \sum_{j=n+1}^{m} \|f_j - 1\|_1$; let $m \to \infty$ and apply (21). A bound $W_n \ge a^n$ is useless as a uniform lower bound when $n \to \infty$. □

If (1) is available, a prescribed summable sequence $\epsilon_j > 0$ can be realized conditionally by taking $\eta_j = \sqrt{1+\epsilon_j} - 1$ and choosing each length after its individual existence threshold. Nothing here covers an arbitrary nonsummable choice of errors, gives the lengths effectively, or requires a companion ultraflat claim. The product statement is secondary; M157's specific addition is the fixed-floor logarithmic budget, not a purported general product-convergence breakthrough.

5.1 Exact finite energy and a physical Gram example

There is a useful finite check on all normalizations. If the stronger condition $d_j > 2\sum_{i<j} d_i(N_i - 1)$ holds, the same last-frequency argument, now with bandwidths doubled, proves

$$\int W_n^2 = \prod_{j\le n} \int f_j^2, \qquad \int W_m W_n = \int W_n^2 \quad (m \ge n). \tag{32}$$

Thus $\|W_m - W_n\|_2^2 = \prod_{j\le m} \int f_j^2 - \prod_{j\le n} \int f_j^2$. Weak separation alone does not justify this squared-bandwidth identity.

For example, $P(z) = 1 + z - z^2$ gives

$$f(t) = 1 - \tfrac{1}{3}(\mathrm{e}(2t) + \mathrm{e}(-2t)), \quad \min f = \tfrac{1}{3}, \quad \max f = \tfrac{5}{3}, \quad \int f^2 = \tfrac{11}{9}.$$

With $d_1, d_2, d_3 = 1, 5, 25$, both identities in (32) are exact: $\int W_3^2 = (11/9)^3$ and $\int W_3 W_2 = (11/9)^2$. This finite example is not a sequence with summable upper excess. Taking all $d_j = 1$ instead already gives $\int f^2 = 11/9 \ne 1$ and shows why normalization cannot be inferred from individual mean one.

For a fixed frequency window $\{-r, \dots, r\}$, its physical Gram matrix in $L^2(W\,dt)$ has entries $G_{k\ell} = \widehat{W}(\ell - k)$, with inner product $\langle u, v\rangle = \int u\bar{v} W\,dt$. If for every $j > m$ the gap $d_j - \sum_{i<j} d_i(N_i - 1) > 2r$, no tail term can change these Fourier coefficients, so $G(W) = G(W_m)$ exactly. The stronger separation ensures this eventually for each fixed r. Consequently

$$a^m I \le G(W) = G(W_m) \le \Big(\prod_{j\le m} b_j\Big) I.$$

These are finite-window margins, not a uniform all-window lower bound. They use the actual weight and actual exponentials, not an arbitrary positive matrix asserted to be a dynamical Gram matrix.

6 Private proof comparison and nonclaims

The fresh private reads distinguish an original derivation from a later comparison. In TR248, lines 142–200 specify the physical weighted Hilbert setting; lines 398–457 prove a finite-frame perturbation bound and track the decoder/quadrature measure; lines 1151–1206 develop moment Gram entries and a finite-interval Taylor remainder. The elementary frame estimate is worth retaining in its correct domain: if

$$\Big\|\sum c_j \phi_j\Big\|^2 \ge A \sum |c_j|^2, \qquad \Big\|\sum c_j(\psi_j - \phi_j)\Big\| \le \rho\, \|c\|_2\,,$$

then the triangle inequality gives lower bound $(\sqrt{A} - \rho)^2 \|c\|_2^2$ when $\rho < \sqrt{A}$. The measure in both norms must be the same. The finite Fourier window above is a concrete setting where that hypothesis can be checked; a positive zeroth moment alone is insufficient.

TR-item-169 is a later corrected comparison, not substituted for that original proof. Its read portions distinguish finite physical frames, summable logarithmic control, uniform integrability

and cyclicity. Proposition 5.1 supplies log integrability and mass for its own explicitly normalized products, but supplies neither a transformation nor cyclicity of an observable. Even a Haar-equivalent measure is not itself a solution of a simple-Lebesgue Koopman-spectrum problem. For instance the positive-coefficient tower polynomial $r^{-1/2}\sum_{j=1}^{r} z^{a_j}$ has value $\sqrt{r}$ at one, regardless of its exponents. Signed Littlewood factors cannot simply be identified with these positive tower factors. The private reads also retain the finite-mass and moving-window compatibility obligations. No whole-corpus reading, tower construction, or Banach-problem solution is claimed.

7 Current public comparison and unresolved compatibility

The search cut-off is October 8, 2026; queries contained only public titles and public mathematical terms. Balister et al. [2] establish flat Littlewood polynomials between two fixed absolute positive multiples of $\sqrt{N}$. That published theorem is not the near-unit upper constant of (1). Its abstract and theorem statement, not its full proof, were checked here. Lovett–Meka's Theorem 4 [3] was checked at the level of its finite-dimensional threshold and starting-point premises; its random-walk proof was not independently audited. The primary Alon–Yuster paper [4] was consulted for the hypergraph matching context, not to replace the source's entire packing argument by a citation.

Erdős's maximum-modulus problem remains relevant to scope. Erdélyi's August 2026 preprint [5], Theorem 2.1, states the lower bound $N + (N-1)^{1/3}/38$ for the squared maximum of a length-N Littlewood polynomial. This is compatible with a fixed upper factor $1+\eta$ at sufficiently large N: it imposes $(1+\eta)^2 - 1 \geq (N-1)^{1/3}/(38N)$, not a fixed positive relative gap. The theorem statement was read; no proof certification is claimed.

There is also an unresolved conflict in public claims, not merely a difference of normalization. El Abdalaoui's preprint [6] defines L^α flatness by $\left\| |P|/\sqrt{N} - 1 \right\|_\alpha \to 0$ and states nonexistence for $\alpha \geq 4$ in its Theorem 1 and Corollary 1. In contrast, (1), along a sequence $\eta \downarrow 0$, would imply exactly such L^4 flatness: (22) gives $\int (f-1)^2 \to 0$, and

$$|\sqrt{f} - 1|^4 \leq (f-1)^2$$

pointwise, since $|\sqrt{f} - 1| \leq \sqrt{f} + 1$. Thus these statements cannot both hold as stated. The definitions and statements were checked, but the competing full proofs were not adjudicated. A further September 2025 preprint [7], consulted at abstract level, makes related nonflatness claims with additional coefficient conditions. We do not resolve this literature conflict by publisher provenance, finite experiments, or the local estimates proved here. All global uses of the M157 headline remain attributed and conditional; Theorem 2.1 is a local analytic statement under its explicit hypotheses.

8 Advantages, costs, and further work

The acute-angle argument changes only the taper width and the treatment of two nonnegative amplitudes. It preserves the source's derivative slots, Hermitian reality, coefficient scale and consecutive degree range. The improvement is therefore directly usable at its correction interface. Its cost is a small geometry-dependent prefactor and a threshold ensuring all strips fit; neither is uniform in the later $\delta \to 0$ limit. The defect estimate is optimal for the available moment information, but exploiting correlations between Y and its particular Fourier correction could do better than the worst-case information class.

The fixed-floor log bounds quantify how near-unit upper control limits deep troughs and Mahler-measure loss, without denying that small exceptional troughs may remain. Their cost is a constant

growing like $|\log a|$ as the floor degenerates. Summability and exact frequency separation are additional product hypotheses, not consequences of a single-length envelope. Finite Fourier windows can be stable while their best lower margins tend to zero with the window size.

Useful further work is consequently specific: expose computable gap and phase constants upstream to make the local $N^{-1/2}$ certificate numerical; investigate the actual joint distribution of sampled coefficients and corrections to improve the rounding budget; and independently reconcile the conflicting global flatness proofs. Higher phase matching or a different exterior projection might improve other rates, but is not proved here. No transform alone supplies the matching, rounding, all-length construction or physical dynamical realization.

9 Verification and contribution boundary

The owned exact-check script uses rational arithmetic and finite Laurent polynomials to test the acute-vector identity, the second integration by parts algebra, taper exponent bookkeeping, sharp rescaling extremizers, endpoint moment budgets, logarithmic bounds with rational enclosures, autocorrelation normalization and separated finite products. It does not construct the source's asymptotic family or simulate a Koopman operator. The author run passed 23,947 exact assertions, including 718 rational logarithmic-enclosure checks. The check JSON records these independently executed assertions, separate from parent-run checks. Rendered-page review, native compiler outcome and export logs are recorded in owned evidence files.

The public source read covers all 1,923 scientific-section lines and 333 additional source/metadata lines in 11 files. Private reading covers 475 unique lines of the later comparison and 175 unique lines of the original TR248, with the complements explicitly recorded. This is not all 584 or all 1,302 lines of those private files, much less the corpus. External literature reading is statement- or abstract-scoped as specified above. Source hashes establish identity continuity, not mathematical truth. All proofs of our local increments are in Theorem 2.1, Proposition 3.1, and their stated consequences; upstream and global obligations remain visible.

C3 is assigned for the meaningful fixed-geometry improvement from $N^{-1/4}$ to $N^{-1/2}$ and the sharp smaller rescaling budget, not for renaming Jensen, Parseval, convexity or Riesz products. These local results are neither a new global existence proof nor a novelty claim against the entire literature. Applicability L3 is unchanged and logically separate.

References

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[3] S. Lovett and R. Meka, *Constructive discrepancy minimization by walking on the edges*, arXiv:1203.5747v2, Theorem 4. Primary preprint.

[4] N. Alon and R. Yuster, *On a hypergraph matching problem*, Graphs and Combinatorics 21 (2005), 377–384. Author-hosted paper.

[5] T. Erdélyi, *On an Erdős Problem about the Maximum Modulus of Littlewood Polynomials on the Unit Circle*, arXiv:2608.00744, 2026, Theorem 2.1. Primary preprint.

[6] E. H. El Abdalaoui, *On L^α-flatness of Erdős–Littlewood's polynomials*, arXiv:2504.21499, Theorem 1 and Corollary 1. Primary version consulted.

[7] E. H. El Abdalaoui, *A generalization of Littlewood's L^α flat theorem, $\alpha > 0$*, arXiv:2509.04212, 2025. Primary abstract consulted.

[8] M. Subbarao, original private report TR248, `TR-rev-1-old-tr248.tex`, selected proof passages as recorded in Section 6. Confidential local material; not uploaded.

[9] M. Subbarao, private corrected comparison `TR-item-169-astra.tex`, selected passages specified in the read ledger. Confidential local material; not uploaded.